Enhancing Best-Selling Books Analysis: Query Optimization and Genre Classification with Deep Learning

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***Abstract*—The rapid digital transformation of the publishing and retail industries has led to the increased use of data-driven approaches for market analysis, customer engagement, and operational efficiency. This paper addresses two interconnected tasks aimed at improving the analysis and classification of book-related data: optimizing SQL queries for structured metadata and classifying book genres using visual cues from cover images.**

The first task focuses on enhancing query performance for a dataset of best-selling books, with a specific emphasis on identifying inefficiencies in query design and execution. Techniques such as replacing correlated subqueries with joins, creating indices on frequently queried columns, and leveraging materialized views were employed. These optimizations resulted in significant performance improvements, with execution times reduced by up to 95%.

The second task leverages deep learning to classify book genres from cover images in the BookCover30 dataset. Using transfer learning with a pre-trained ResNet50 model, the classification pipeline included data augmentation, weighted loss functions, and dynamic learning rate scheduling to address challenges such as class imbalance and overfitting. While achieving a modest accuracy of 12%, the results exceeded random guessing, highlighting the model's ability to distinguish certain genres based on unique visual patterns.

***Index Terms*—SQL query optimization, deep learning, genre classification, data augmentation, publishing industry, retail analytics, metadata analysis, transfer learning, database indexing, image-based prediction**

I. INTRODUCTION

The digital transformation of the publishing and retail industries has led to an increasing reliance on data-driven strategies to enhance operations, customer engagement, and market analytics. This project explores two distinct yet interconnected tasks that address complementary aspects of book-related data: (1) optimizing query performance in structured metadata and (2) classifying book genres using visual cues from cover images.

The first task focuses on improving the performance of complex SQL queries executed on a structured dataset of best-selling books. This dataset, containing information about authors, titles, sales, and more, but contains many missing values in a critical column—genre. Efficient query execution and optimization are paramount in database systems, particularly for queries involving large datasets or multiple joins. By addressing inefficiencies in query design, such as correlated subqueries and redundant scans, this task aims to reduce execution times significantly. This work involves both logical and physical query optimization techniques, including the use of indexing, join order reorganization, and query plan evaluation to identify and implement the most cost-effective execution strategies.

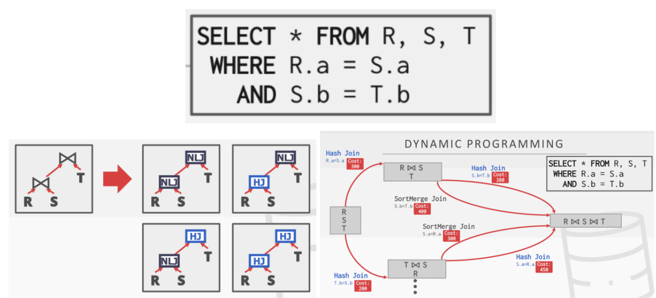


Fig. 1: Example of Physical Query Optimization Task.

The diagram illustrates query optimization for a join query involving R, S, and T. Different join methods (e.g., Nested Loop, Hash Join) and orderings are evaluated, with dynamic programming used to compute and select the lowest-cost execution plan. Source: *Database Management Systems* by Raghu Ramakrishnan and Johannes Gehrke. [4]

The second task extends beyond structured metadata to tackle the classification of book genres based on unstructured visual data—book cover images. Utilizing the BookCover30 dataset, which contains approximately 57,000 book covers spanning 30 genres, this task leverages a pre-trained ImageNet model for deep learning. By fine-tuning the ResNet50 architecture, the project aims to accurately predict the genre of a book from its cover image. This approach highlights the potential of transfer learning to adapt general-purpose models to domain-specific tasks.

The two tasks in this project—query optimization for structured metadata and genre classification from book cover images—are connected through the shared goal of addressing the missing values in the genre column in the best-selling books dataset. By leveraging genre predictions from the second task, the project integrates structured metadata with unstructured visual data to enable a more comprehensive analysis of books.

By synthesizing these advancements in database optimization and deep learning, this project offers a scalable, multifaceted solution to challenges in data analysis and prediction, with significant applications in publishing and retail. For example, these tasks offer practical applications for publishing and e-commerce, enabling rapid insights into market trends and customer preferences while supporting personalized recommendations and competitive analysis. By integrating database optimization strategies with deep learning, the project highlights the connection between metadata and visual data in addressing modern data challenges efficiently and effectively.

II. Related Works

A. Study 1: Query Optimization [1]

Query optimization in database management systems (DBMSs) involves selecting the most efficient access plan to process a query, as different plans can vary significantly in execution cost. The query optimizer comprises modules like the Rewriter, which transforms queries into more efficient forms, and the Planner, which uses strategies such as dynamic programming to explore access plans and select the least costly one. Supporting components include the Algebraic Space, which orders operators, the Method-Structure Space, which specifies execution methods, the Cost Model, which estimates plan costs, and the Size-Distribution Estimator, which predicts result sizes. While query optimization has been extensively studied, challenges remain, especially for advanced queries in areas like distributed and parallel systems, highlighting the field's ongoing relevance and complexity.

B. Study 2: ImageNet: A Large-Scale Hierarchical Image Database [2]

The ImageNet project introduced a large-scale, hierarchical image database designed to address challenges in organizing and utilizing the explosion of online image data. Built on the semantic structure of WordNet, ImageNet features millions of high-resolution, annotated images across thousands of categories, with plans to expand to 50 million images. The database emphasizes clean annotations, diverse content, and a densely populated hierarchy, making it a valuable resource for object recognition, classification, and clustering. Its construction relied on innovative methods like leveraging Amazon Mechanical Turk for labeling. ImageNet's scale, accuracy, and semantic richness set it apart from other datasets, offering unparalleled opportunities for advancements in computer vision and related fields.

C. Study 3: Are We Done with ImageNet? [3]

ImageNet's labeling approach has significant limitations, including assigning a single label to images with multiple objects, overly restrictive label proposals, and arbitrary class distinctions, which result in biased and noisy data. To address these issues, researchers proposed Reassessed Labels (ReaL) that better capture image diversity, multi-label training objectives, and a cleaned dataset to reduce noise and improve generalization. These enhancements, along with the new ReaL accuracy metric (instead of evaluating a model based on a single "correct" label per image, ReaL accuracy measures whether the model's top-1 prediction is included in a set of acceptable labels derived from diverse human annotations), provide a more meaningful evaluation of model performance, highlighting that recent models outperform original ImageNet labels and prompting the need for evolving benchmarks as models near-optimal performance.

III. Method

A. Task 1: Query Optimization Analysis

The first task focuses on optimizing SQL queries to efficiently retrieve metadata from a dataset of best-selling books. Using `SQLite3`, a lightweight database engine, the dataset was managed in memory for rapid execution. Indices were created on frequently queried columns such as `Genre` and `Author` to reduce search time and enable faster lookups and joins. Key queries, such as (1) identifying authors who have written across diverse genres, such as both "Children's" and "Fiction" (2) identifying authors who wrote in diverse genres or those spanning all genres written by J.K. Rowling, were optimized by replacing correlated subqueries with joins and leveraging materialized views to minimize redundant computations.

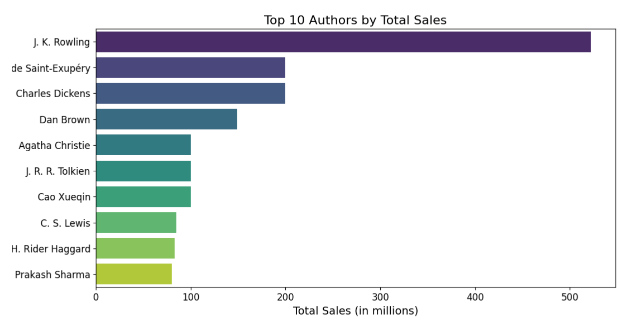


Fig. 2: Top 10 Authors by Total Sales.

J.K. Rowling leads the list with significantly higher sales than the other authors in this dataset, likely due to the global success of the Harry Potter series. This insight could be valuable for forming queries, such as identifying authors who write in similar genres. Such data could help a publishing company strategize to replicate Rowling's success.

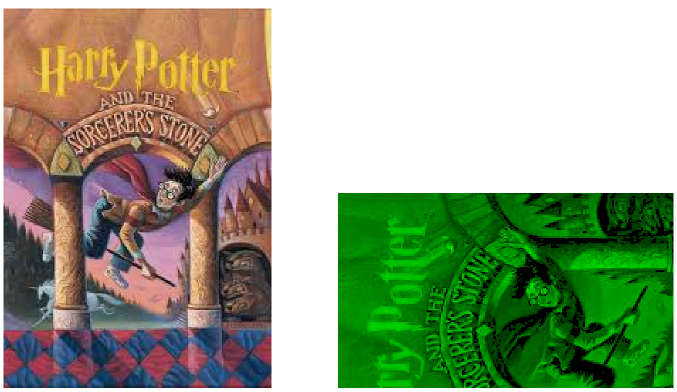
A notable feature of the methodology was the use of Python's `time.perf_counter()` to measure query execution times with high precision, enabling iterative performance improvements. To handle query results efficiently, a custom iterator class was developed to process outputs in batches, ensuring scalability and memory efficiency even with large datasets. Additionally, the EXPLAIN QUERY PLAN tool was used to analyze and refine execution plans, providing insights into costly operations and guiding optimizations. These techniques collectively demonstrate a comprehensive approach to improving query performance and scalability in real-world applications.

B. Task 2: Genre Classification with Deep Learning

The second task addresses classifying book genres based on cover images from the BookCover30 dataset, which includes genre annotations. The task aims to develop a deep learning model capable of predicting genres for unseen covers, leveraging a pre-trained `ResNet50` model fine-tuned through transfer learning. The model's final fully connected layer is adapted to the number of genres in the dataset, ensuring compatibility with the classification task.

To prepare the dataset, image links were converted into downloaded files using a script, with error-handling mechanisms to skip broken or inaccessible links. This ensured a consistent dataset by excluding invalid images and logging errors for review. Key preprocessing steps included:

- Data augmentation techniques such as random rotations, cropping, and color jittering, which enhance generalization by exposing the model to diverse variations in the data



(a) Original book cover. (b) New image with data augmentation.

Fig. 3: Example of random rotations, cropping, and color jittering (Disclaimer: Only the image data is transformed; the file itself remains unmodified)

- To tackle class imbalance, a weighted cross-entropy loss function was employed, prioritizing underrepresented genres during training
- The training strategy combined layer freezing, allowing pre-trained features to guide initial learning, with selective unfreezing for domain-specific adaptation.
- A learning rate scheduler dynamically adjusted the optimization process, contributing to improved training stability and convergence.

Together, these strategies enabled robust genre classification with scalable and error-resilient preprocessing.

IV. Results

A. Task 1: Query Optimization Analysis

Two key queries were optimized to address performance bottlenecks and logical inefficiencies. The first query identified authors who wrote both ”Children's” and ”Fiction” books, while the second query found authors whose works span all genres written by J.K. Rowling.

*1) **Query Logic and Execution Plan**:* Initially, the queries relied heavily on correlated subqueries, which performed row-by-row comparisons and caused significant performance overhead due to multiple nested scans. For example, in Query 1, the correlated subquery executed a separate scan for each row in the dataset. By leveraging joins instead of correlated subqueries, the logic was streamlined, enabling the database engine to process data more efficiently.

The `EXPLAIN QUERY PLAN` feature in SQLite was used to analyze execution plans and identify inefficiencies. The analysis revealed redundant operations such as full-table scans. To address these, indices were created on frequently queried columns like `Genre` and `Author(s)`. These indices enabled faster lookups by reducing the cost of searching and sorting operations. For instance, in Query 2, a materialized table for J.K. Rowling's genres replaced a dynamic subquery, further optimizing performance.

*2) **Custom Iterator for Scalable Processing**:* To handle large query outputs efficiently, a custom iterator class was implemented. This class processed results in small, manageable batches using the `fetchmany` method, which avoided memory overflow and allowed for scalable data processing. The iterator enabled the seamless handling of queries even with increasing dataset sizes, ensuring consistent performance.

*3) **Performance Improvements**:* The optimizations resulted in significant performance gains:

Fig. 4: Execution Times Before and After Optimization. Disclaimer: Execution times may vary depending on hardware and other conditions.

- Query 1: Execution time decreased from 0.0083 seconds to 0.0004 seconds by replacing correlated subqueries with joins and adding indices.
- Query 2: Execution time decreased from 0.0078 seconds to 0.0005 seconds through the use of materialized tables, indices, and efficient query logic.

B. Task 2: Genre Classification with Deep Learning

Despite a modest overall accuracy of 12%, the results exceeded random guessing (3.33% for 30 genres). This experiment was conducted using a small subset of the dataset, and running the model required downloading the images used in both the training and testing datasets. Notable points include:

- **Overfitting Evidence:** Rapid loss reduction and a large gap between training and validation performance suggest potential overfitting. Data augmentation and weighted loss partially mitigated class imbalance, but further regularization strategies could enhance generalization.

```
Epoch 1, Loss: 5444.6295
Epoch 2, Loss: 5181.0997
Epoch 3, Loss: 5002.3438
Epoch 4, Loss: 4870.6445
Epoch 5, Loss: 4707.0129
Epoch 6, Loss: 4428.0339
Epoch 7, Loss: 3977.0406
Epoch 8, Loss: 3361.3659
Epoch 9, Loss: 2582.1781
Epoch 10, Loss: 1816.8305
```

```
Epoch 11, Loss: 905.7437
Epoch 12, Loss: 713.5283
Epoch 13, Loss: 532.4463
Epoch 14, Loss: 417.9893
Epoch 15, Loss: 351.5671
Epoch 16, Loss: 292.1248
Epoch 17, Loss: 270.9076
Epoch 18, Loss: 251.1580
Epoch 19, Loss: 260.5567
Epoch 20, Loss: 257.7206
```

Fig. 5: Cross Entropy Loss Across Epochs.

- **Insights from Metrics:** Some genres, like "Children's," achieved higher precision and recall, likely due to distinct visual patterns in their covers, while others performed poorly due to overlapping design elements or insufficient data representation.
- **Visualizing Results:** The confusion matrix revealed misclassifications between similar genres, underscoring the challenge of visually similar covers.

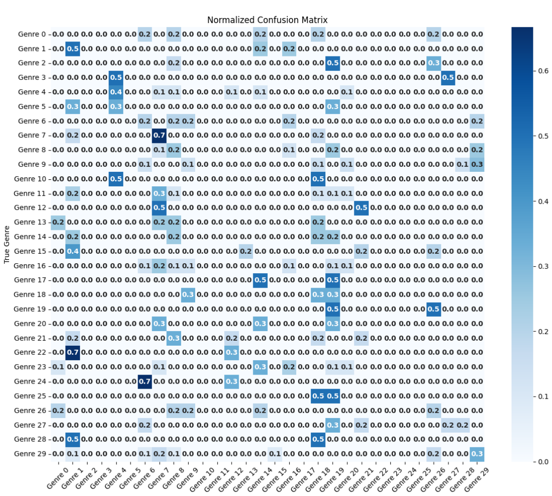


Fig. 6: Confusion Matrix

The confusion matrix for task 2 reveals key insights into the model's genre classification performance, highlighting both strengths and weaknesses. Notably, genre 24 ("Science Fiction & Fantasy") was frequently mistaken for genre 6 ("Computers & Technology") with a normalized value of 0.7. This misclassification suggests overlapping visual features between these genres, possibly due to futuristic or technical imagery on book covers. Similarly, genre 22 ("Romance") was commonly predicted as genre 1 ("Biographies & Memoirs") with a value of 0.7, indicating that romantic themes or cover designs may share elements with biographical aesthetics.

On the positive side, the model accurately predicted genre 7 ("Cookbooks, Food & Wine") with a high confidence value of 0.7, showcasing its ability to identify visually distinct patterns such as food imagery or culinary designs. However, genre 25 ("Self-Help") exhibited significant ambiguity, with an even 50-50 split in predicted labels between genre 18 ("Parenting & Relationships") and genre 19 ("Politics & Social Sciences"). This suggests that self-help covers may lack unique features, often resembling covers from related genres.

These patterns underscore the challenges of classifying visually similar genres and the need for additional features or improvements in model generalization to reduce cross-genre confusion. Future enhancements might include incorporating text analysis or additional data augmentation techniques to better capture subtle differences in cover designs. This aligns with observations in *Are we done with ImageNet?* [3], which highlighted the limitations of single-label assignments for images containing multiple objects and the noise introduced by restrictive class distinctions. The study's proposal of Reassessed Labels (ReaL) to account for image diversity and multi-label training objectives underscores the importance of evolving benchmarks and evaluation metrics. Applying similar principles to this project, such as multi-label objectives or better dataset cleaning, could significantly improve the model's ability to differentiate between visually overlapping genres.

V. Conclusion

In conclusion, this project demonstrated the integration of database query optimization techniques and deep learning-based genre classification to address challenges in analyzing book-related data. Task 1 achieved significant reductions in query execution times by replacing correlated subqueries with joins, employing materialized tables, and leveraging indices on frequently queried fields. The use of SQLite's `EXPLAIN QUERY PLAN` and custom iterators ensured both efficient execution and scalability, highlighting best practices for metadata retrieval in structured datasets. These optimizations enabled rapid insights into authorship patterns and genre diversity with practical implications for publishers and retailers.

Task 2 extended the analysis to unstructured data by developing a deep learning model for genre classification using book cover images. While the model achieved modest accuracy, it surpassed random guessing, demonstrating the feasibility of leveraging visual cues for classification. Challenges, such as cross-genre confusion due to visually overlapping features, underscored the need for improved generalization through additional data augmentation or the incorporation of textual features. Drawing from studies like *Are we done with ImageNet?*, adopting multi-label objectives or reassessing dataset labels could provide more nuanced evaluations and enhance performance.

A significant limitation in Task 2 was the use of only a small subset of the dataset, with 1,600 covers in the training set and 100 in the testing set, compared to the full dataset of 57,000 covers. Using the complete dataset in future work would likely improve the model's performance and generalization by providing a richer and more diverse set of training examples. However, memory usage and computational constraints were critical barriers to utilizing the full dataset in this project. A potential solution to this issue could involve leveraging high-performance computing resources, such as the Great Lakes cluster, to handle larger datasets efficiently. The use of distributed training and batch processing on such platforms could mitigate memory limitations, enabling the training of more robust and scalable models while maintaining efficient resource utilization. These enhancements could pave the way for more impactful applications in the publishing and retail industries.

GitHub Link: https://github.com/ninabryan23/STATS507_Public/tree/main/Final_Project

APPENDIX

TABLE I: Category-wise Training and Test Full Genre Dataset Sizes

Label	Category Name	Training Size	Test Size
0	Arts & Photography	1,710	190
1	Biographies & Memoirs	1,710	190
2	Business & Money	1,710	190
3	Calendars	1,710	190
4	Children's Books	1,710	190
5	Comics & Graphic Novels	1,710	190
6	Computers & Technology	1,710	190
7	Cookbooks, Food & Wine	1,710	190
8	Crafts, Hobbies & Home	1,710	190
9	Christian Books & Bibles	1,710	190
10	Engineering & Transportation	1,710	190
11	Health, Fitness & Dieting	1,710	190
12	History	1,710	190
13	Humor & Entertainment	1,710	190
14	Law	1,710	190
15	Literature & Fiction	1,710	190
16	Medical Books	1,710	190
17	Mystery, Thriller & Suspense	1,710	190
18	Parenting & Relationships	1,710	190
19	Politics & Social Sciences	1,710	190
20	Reference	1,710	190
21	Religion & Spirituality	1,710	190
22	Romance	1,710	190
23	Science & Math	1,710	190
24	Science Fiction & Fantasy	1,710	190
25	Self-Help	1,710	190
26	Sports & Outdoors	1,710	190
27	Teen & Young Adult	1,710	190
28	Test Preparation	1,710	190
29	Travel	1,710	190

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